

Instructions to Learners of BSCCS2005

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1. Introduction

This document contains important information regarding the Java course. We will keep updating this document with additional inputs as and when needed. Please read this document carefully.

2. Installing Java SE 11

JDK (or Java Development Kit) is a collection of tools to develop Java based applications. JRE (or Java Run-time Environment), on the other hand, is a set of components necessary to run Java programs. Whenever we install JDK in our computer, JRE is also installed automatically. If we want to only run the Java applications, then JRE is enough. However, if we want to develop Java applications, then we must install JDK.

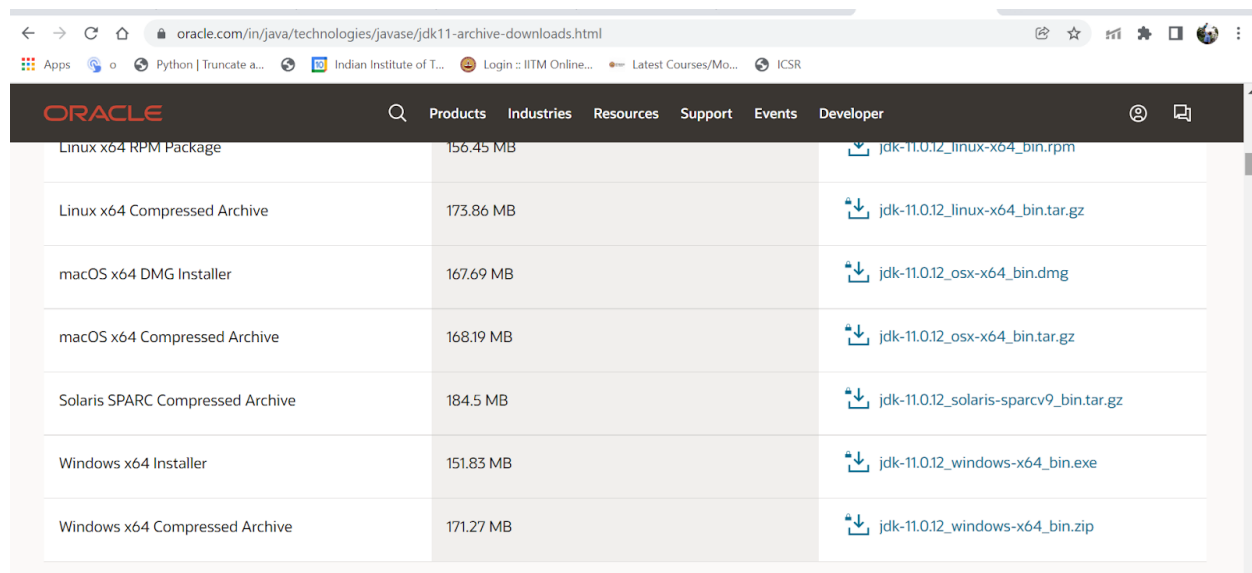
We strongly recommend that you install Java 11 in your computer.

2.1 Installing in Windows

For Windows users, please go through the installation instructions given below.

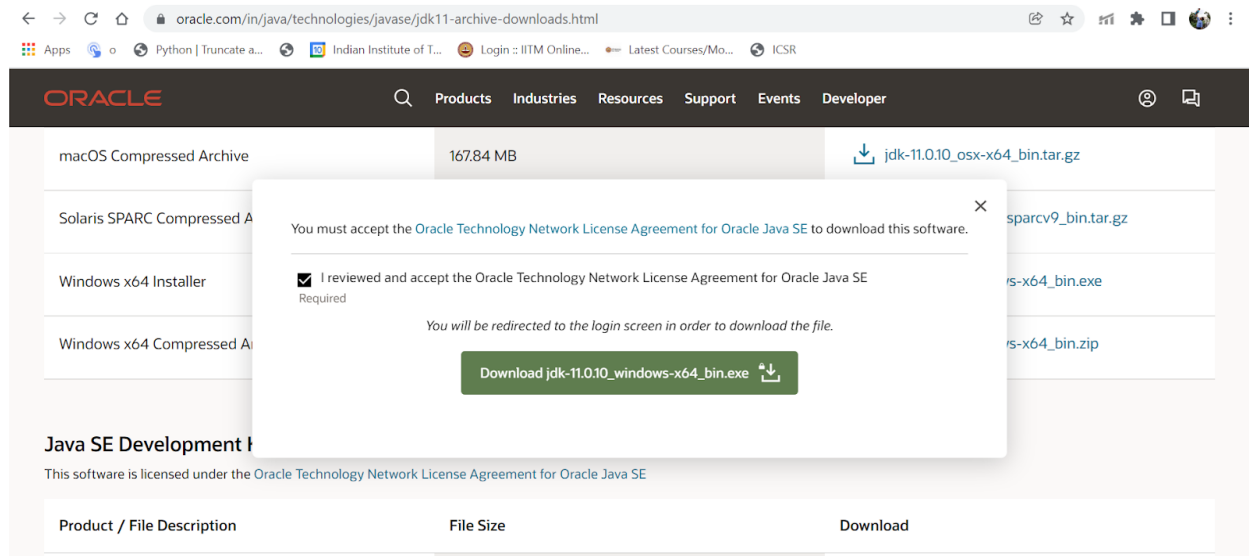
1) Open the URL:

<https://www.oracle.com/in/java/technologies/javase/jdk11-archive-downloads.html>

A screenshot of a web browser displaying the Oracle JDK 11 archive downloads page. The browser's address bar shows the URL: https://www.oracle.com/in/java/technologies/javase/jdk11-archive-downloads.html. The page features a dark navigation bar with the Oracle logo and links for Products, Industries, Resources, Support, Events, and Developer. Below the navigation bar, there is a table listing various download options for JDK-11.0.12 across different operating systems and architectures. Each row includes the download type, the file size, and a direct download link with a download icon.

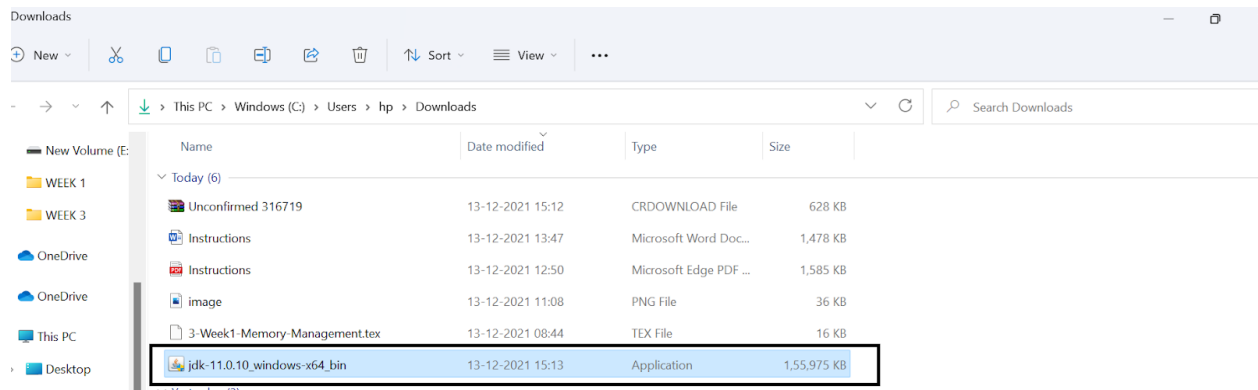
Linux x64 RPM Package	156.45 MB	jdk-11.0.12_linux-x64_bin.rpm
Linux x64 Compressed Archive	173.86 MB	jdk-11.0.12_linux-x64_bin.tar.gz
macOS x64 DMG Installer	167.69 MB	jdk-11.0.12_osx-x64_bin.dmg
macOS x64 Compressed Archive	168.19 MB	jdk-11.0.12_osx-x64_bin.tar.gz
Solaris SPARC Compressed Archive	184.5 MB	jdk-11.0.12_solaris-sparcv9_bin.tar.gz
Windows x64 Installer	151.83 MB	jdk-11.0.12_windows-x64_bin.exe
Windows x64 Compressed Archive	171.27 MB	jdk-11.0.12_windows-x64_bin.zip

2) Click on Windows x64 Installer



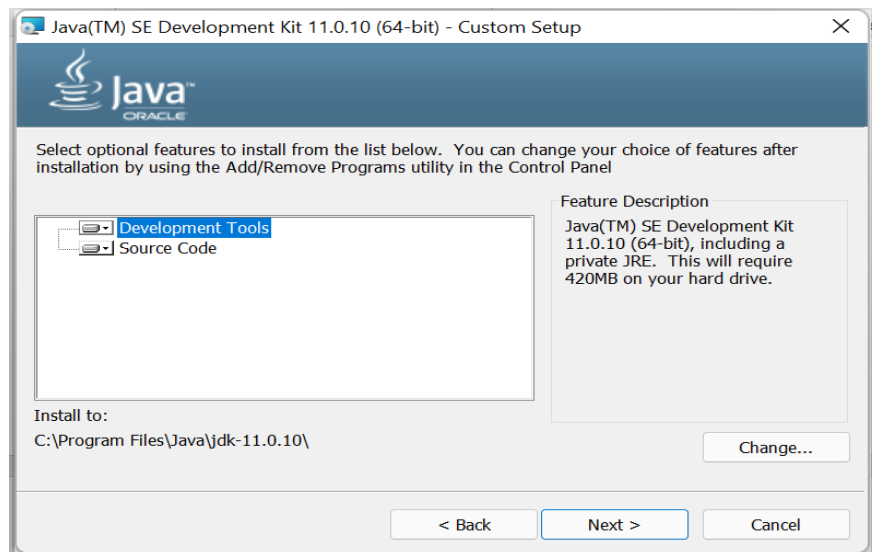
3) Accept the license agreement and click on download.

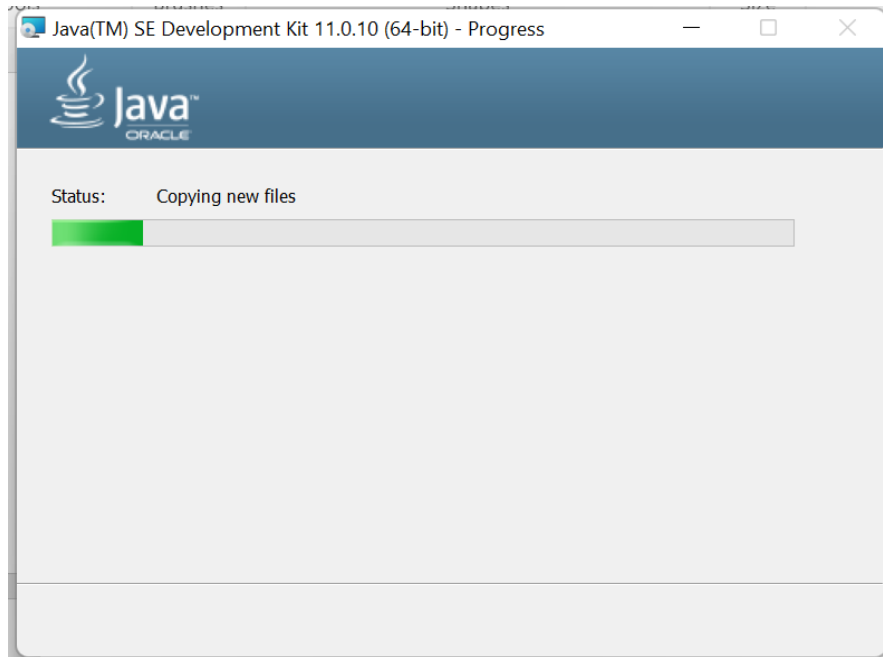
4) If it asks you to create an account, then create the account and proceed with the downloading.



5) Right-click the downloaded file and select Run as administrator.

6) Follow the instructions on the screen to complete the installation. See the screenshots below.



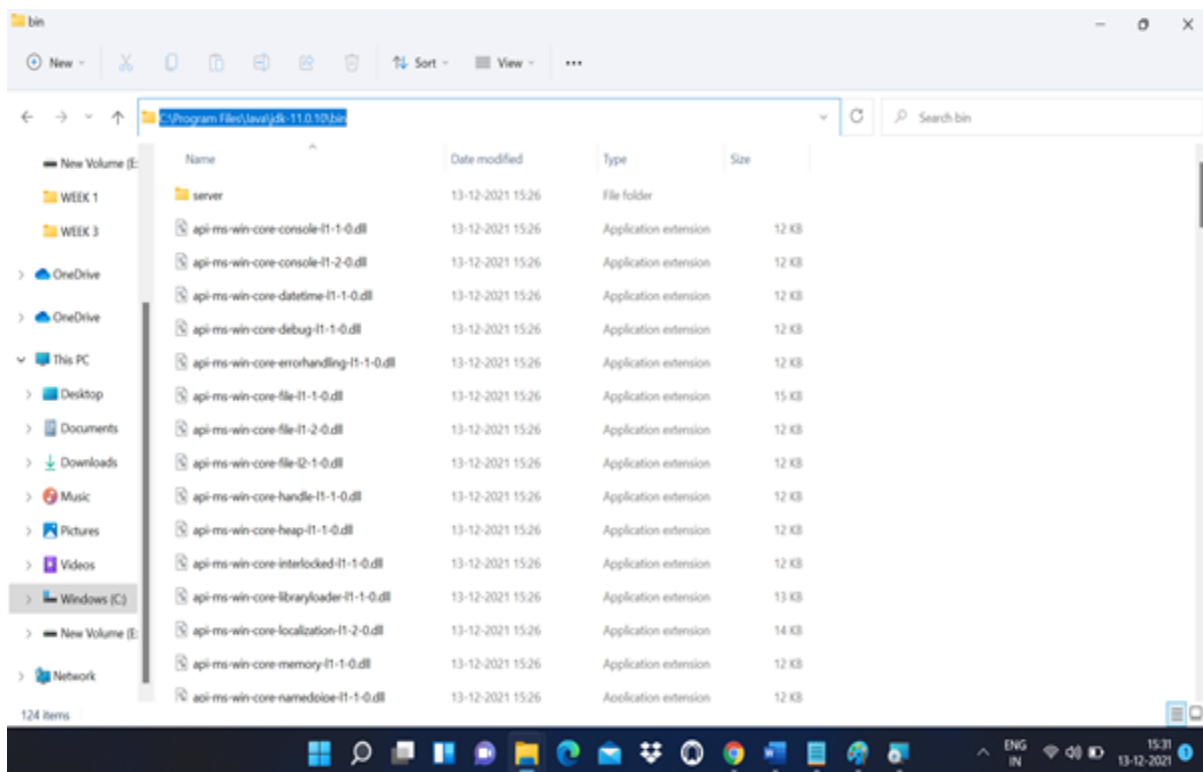


Once the installation is complete, we must set the path variables.

2.2 Setting the path variable

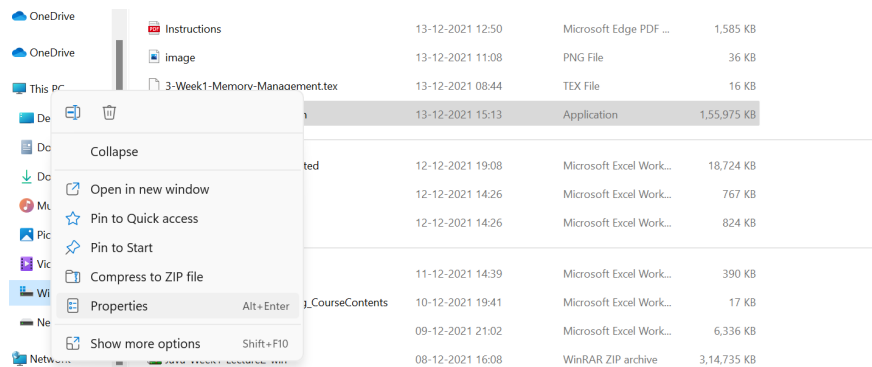
To compile and run our Java programs, we must set the path for JDK. Please follow the steps given below.

- 1) Open the installation directory in your system.
(For ease of explanation, we will assume that the installation directory is C:\Program Files\Java\jdk-11.0.10. Please change it as is appropriate.)

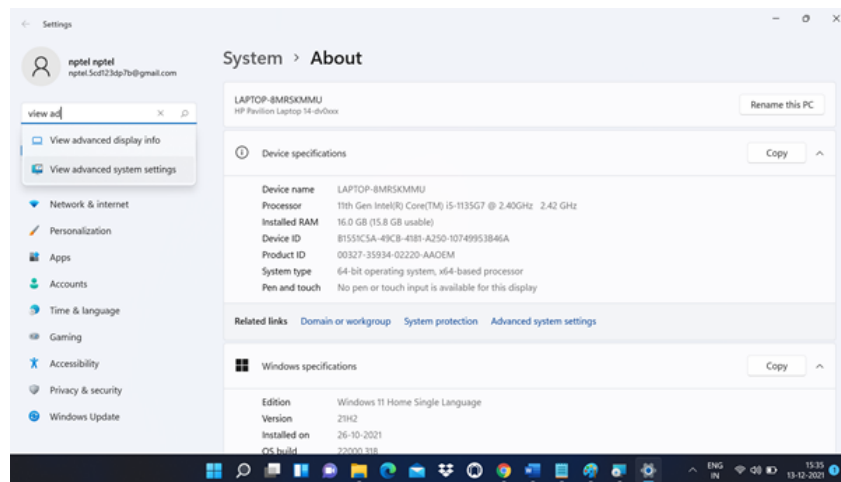


- 2) Copy the path of bin folder inside the installation directory: *C:\Program Files\Java\jdk-11.0.10\bin*.

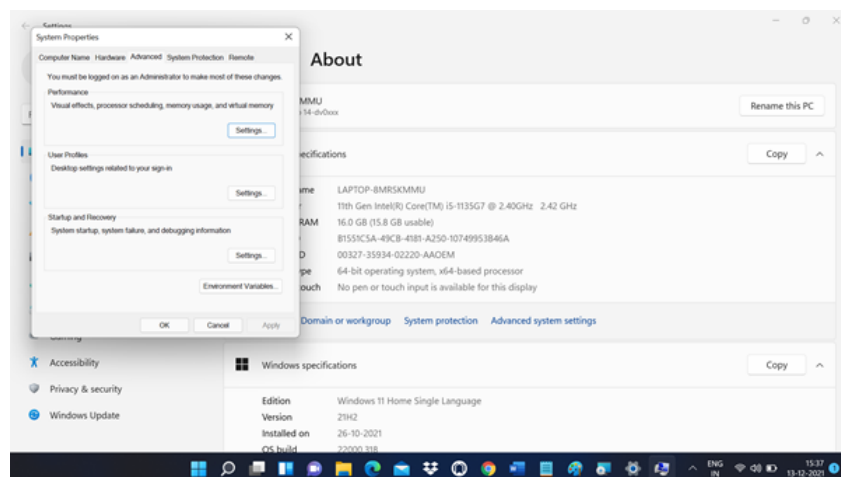
3) Right-click on My computer (This PC) and select properties.

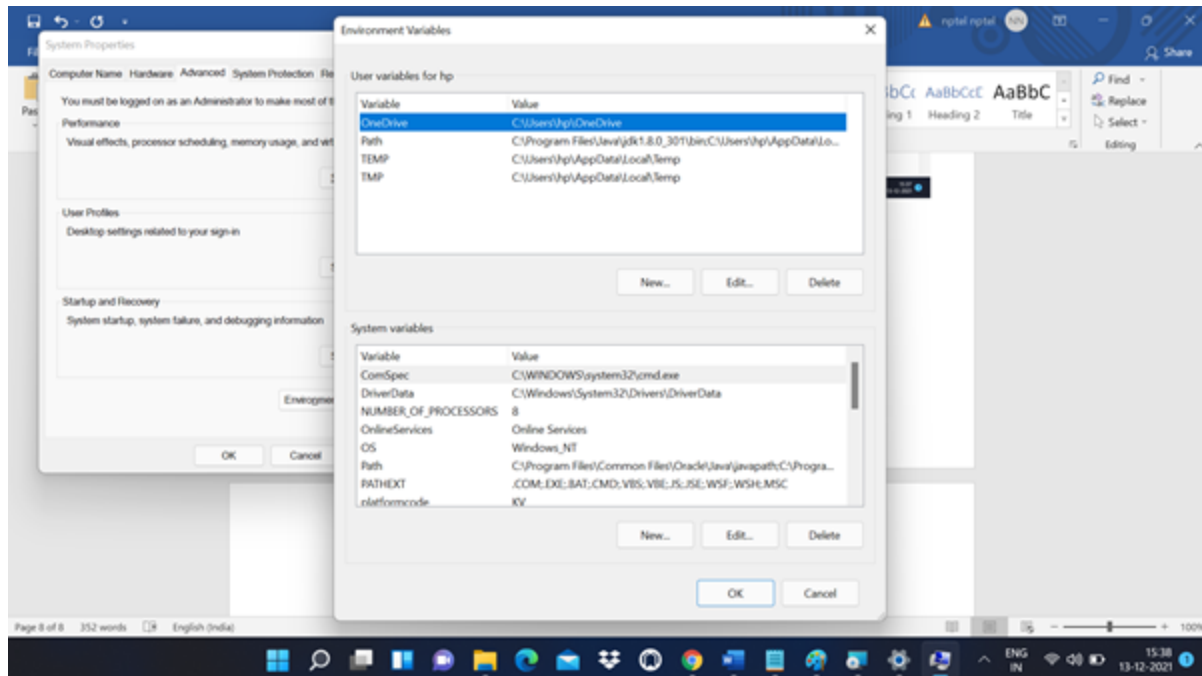


4) Search for "View Advanced System Settings" and click on it.

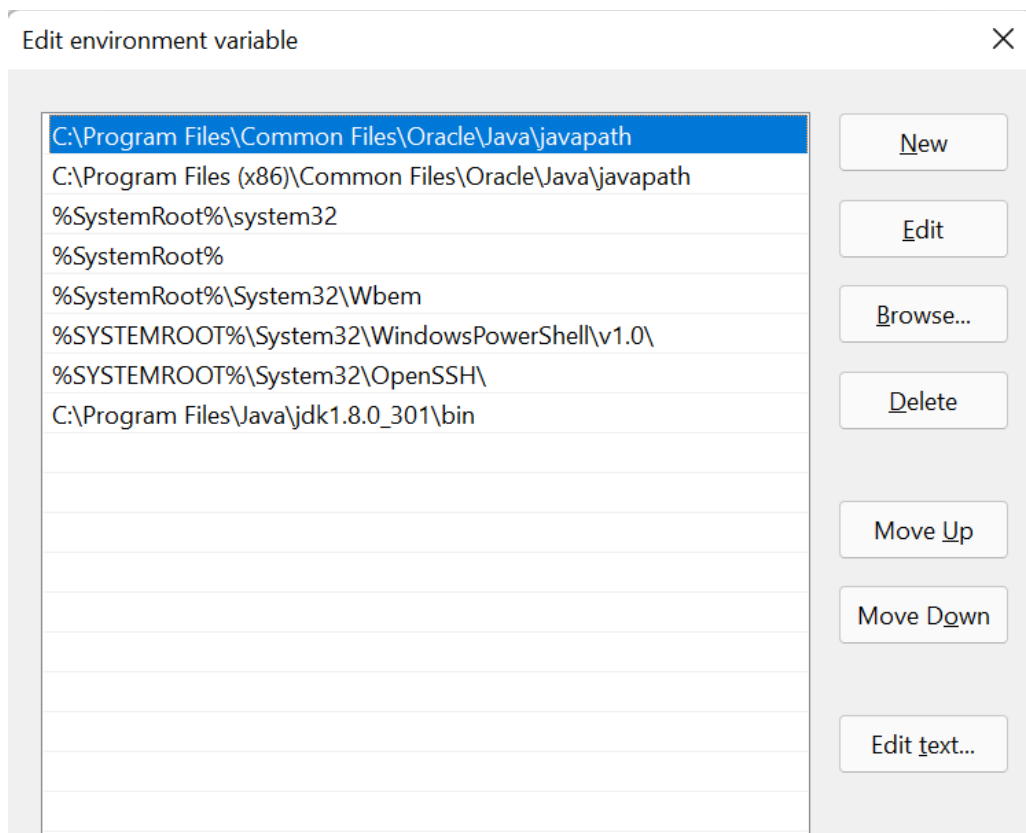


5) Click on Environment variables.





6) Click on "Path" in "User variables". Then, click on "Edit".



- 7) Click on "New" and add "C:\Program Files\Java\jdk-11.0.10\bin" to the list. Click OK.
- 8) Setting up the path variable is complete.

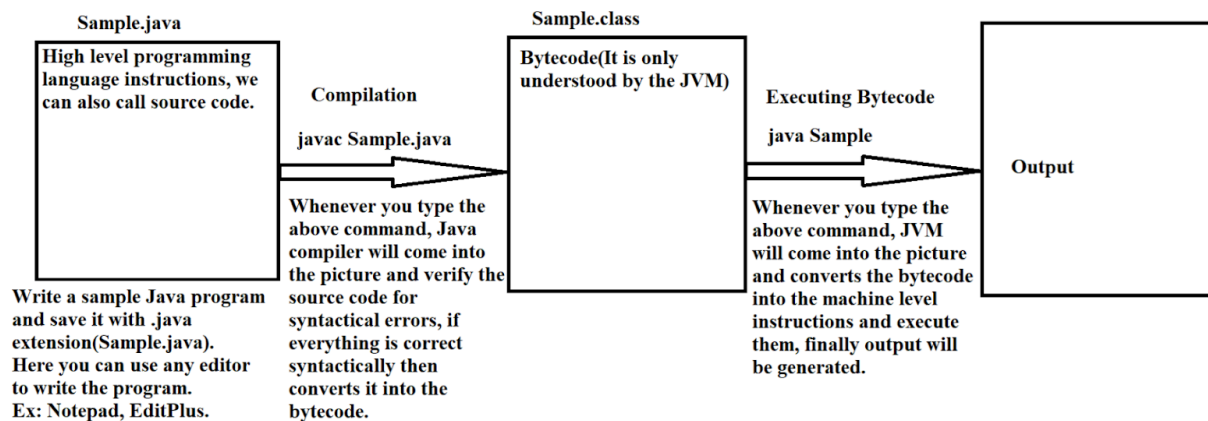
2.3 How to run a Java program?

There are two ways to run a Java program.

- 1) Using command prompt
- 2) Using Eclipse IDE (GUI based)

2.3.1 Using command prompt

Please follow the steps given below to run Java programs from command prompt.



Step1: Open any editor to write the Java program.

Step 2: Save the above written program in your preferred folder, say D:\Java. Let the name of the Java file be Sample.java. Note that the file name should be the same as the class name in the program, and the file must have .java as extension.

Step 3: Now open the command prompt from the file location D:\Java.

Step 4: Compile the Java program using **javac Sample.java** command.

```
D:\Java>javac Sample.java
```

Step 5: Verify that Sample.class file is generated inside D:\Java.

Step 6: Run the java program using **java Sample** command.

```
D:\Java>java Sample
```

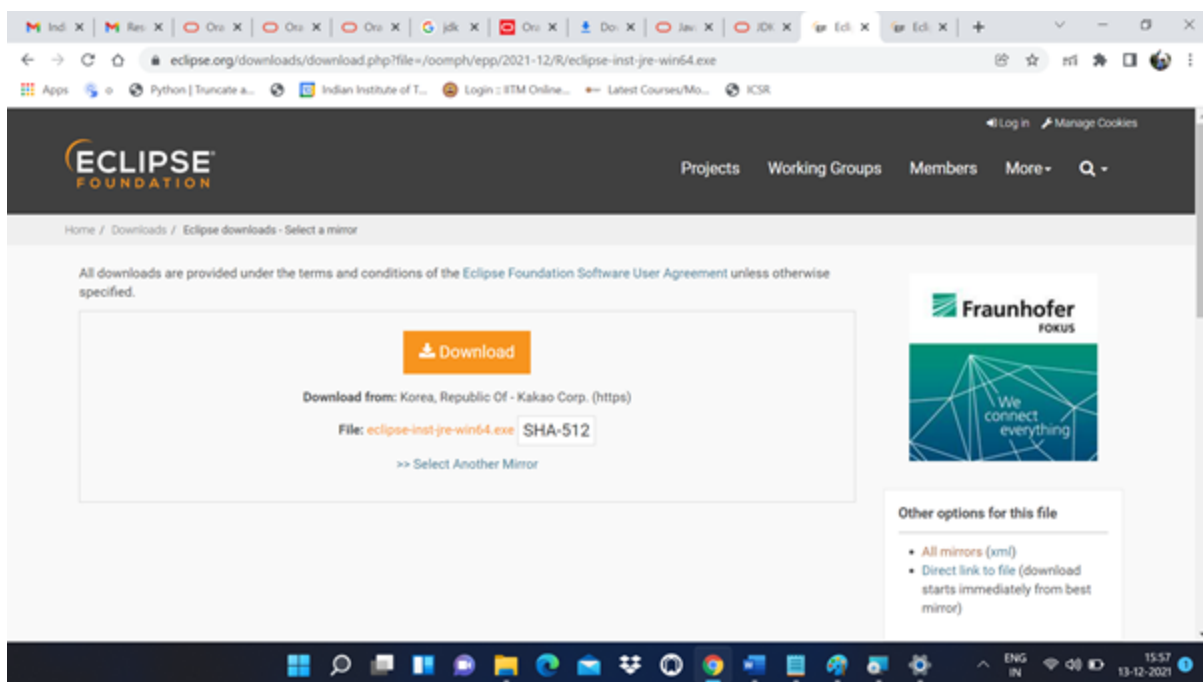
2.3.2 Using Eclipse IDE (GUI based)

Eclipse is an integrated development environment that contains a base workspace and an extensible plug-in system for customizing the environment.

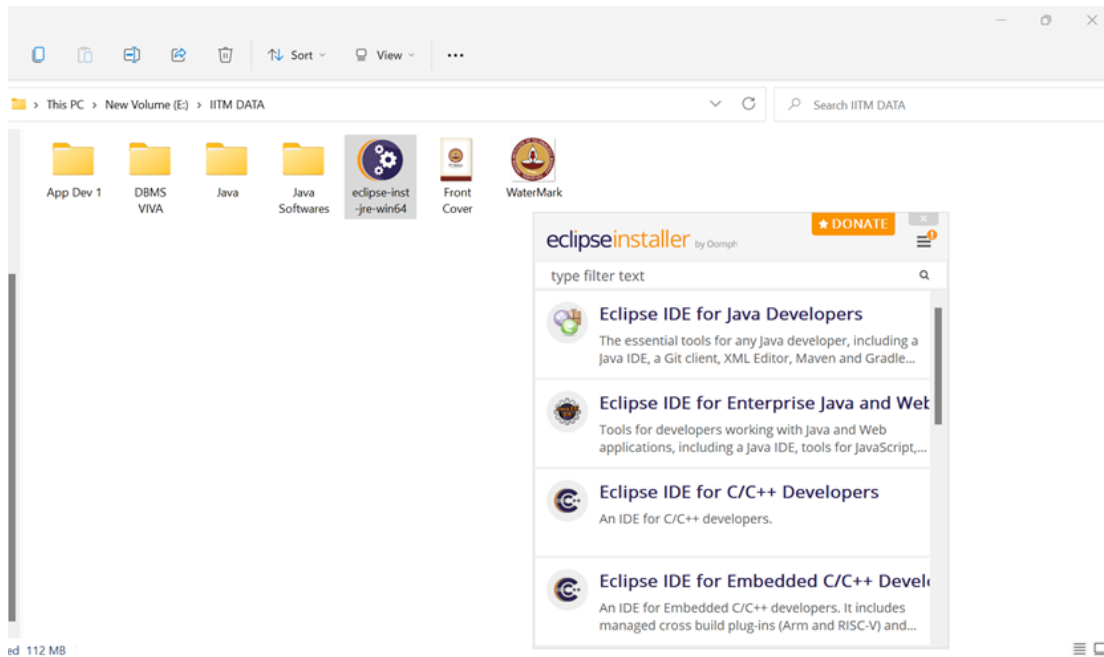
Eclipse IDE is popular for Java Integrated Development Environment (IDE), although the Eclipse environments for other programming languages are also equally popular.

1) Go to the URL:

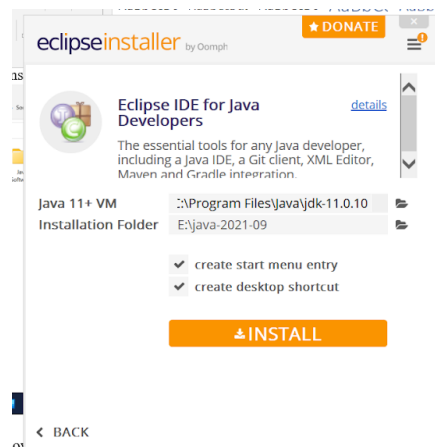
<https://www.eclipse.org/downloads/download.php?file=/oomph/epp/2021-12/R/eclipse-inst-jre-win64.exe>



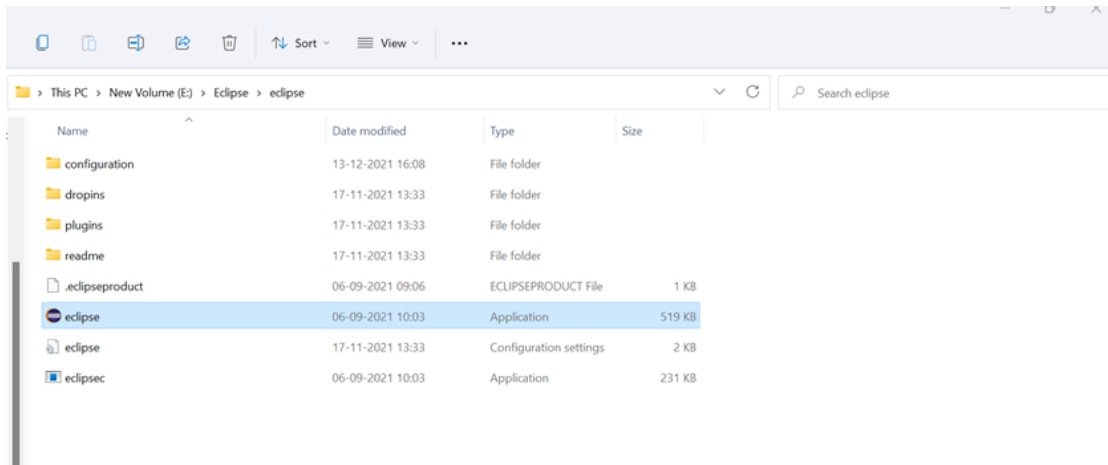
2) Click on *Download* or *eclipse-inst-jre-win64.exe*. You will see the eclipse installer as shown below. Select Eclipse IDE for Java Developers.



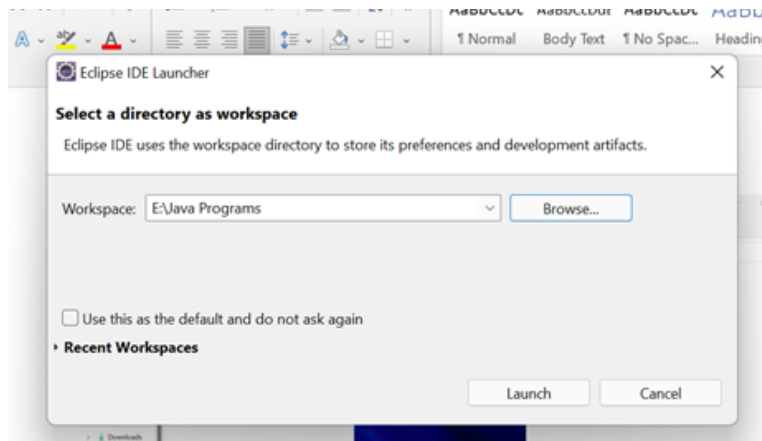
- 3) Eclipse will search for the JDK installation directory. Select the installation folder and click on INSTALL.



- 4) Once the installation is complete, double-click the eclipse Application icon to open the application.



5) Select a directory as workspace, where you want to save your files.



2.4 Resources for installation of Java SE 11 in other operating systems

Installation on Linux Platforms

<https://docs.oracle.com/en/java/javase/11/install/installation-jdk-linux-platforms.html#GUID-737A84E4-2EFF-4D38-8E60-3E29D1B884B8>

Installation on macOS

<https://docs.oracle.com/en/java/javase/11/install/installation-jdk-macos.html#GUID-2FE451B0-9572-4E38-A1A5-568B77B146DE>

2.5 Online Java Compilers:

Even though we strongly recommend that you install Java in your computer as described in earlier sections, it is also important to know that there are many online Java compilers which can be used for practice. We list below the links to a few such compilers.

<https://www.jdoodle.com/online-java-compiler/>

https://www.onlinegdb.com/online_java_compiler

<https://www.programiz.com/java-programming/online-compiler/>

3. Live Sessions

3.1 Activity Sessions

In these sessions, we will address the Activity Questions for the given week. During these sessions, we plan to provide extra support to beginner learners of Java. However, all students are welcome to join.

3.2 Practice/Solve with Instructor Sessions

In these sessions, we will address questions from Practice Assignments. There may be questions for which detailed solutions have not been provided in the solution document. We will pose such questions as 'Solve with the Instructor' questions and lead you through solving those. We request your cooperation in having familiarized yourself with the questions, before coming to the session so that we can make the best utilization of the 2 hours.

3.3 Open Sessions

You are free to ask any questions from the week, except from the graded assignments whose submission deadlines are not yet over.

3.4 Revision Sessions

These will be planned as and when required.